

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

CFD2D Solver — Agentic Benchmark Submission

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Abstract

This report documents a cell-centered finite-volume solver for the two-dimensional compressible Navier–Stokes equations on unstructured meshes composed of triangles and quadrilaterals. The solver is implemented in C++17 with MPI parallelism; domain decomposition uses METIS graph partitioning with non-blocking halo exchange via `MPI_Isend/MPI_Irecv`. Spatial discretization employs the Rusanov (local Lax–Friedrichs) approximate Riemann solver with second-order piecewise-linear reconstruction based on least-squares gradients and a Venkatakrishnan limiter, supplemented by a face-based minmod limiter for additional TVD-like stability. Steady cases use a symmetric Gauss–Seidel (SGS) implicit relaxation with an enhanced diagonal term and a CFL ramp capped at 5. The transient cylinder Re 200 case uses second-order backward-differentiation (BDF2) physical-time integration with point-implicit inner iterations. Meshes are read from multi-zone CGNS files with automatic zone merging.

Three inviscid NACA 0012 cases were completed: Mach 0.15 (2.4 orders residual reduction, 20 000 steps), Mach 0.80 (4.0 orders, converged at 11 898 steps), and Mach 2.0 (1.9 orders, 3 774 steps of 40 000 maximum). The transient cylinder Re 200 case reached $t = 64.9$ of $t_{\text{final}} = 300$ before the wall-clock budget expired; it exhibits monotonically decreasing drag and near-constant lift at that stage, indicating the flow has not yet reached periodic vortex shedding. NACA 0012 laminar Re 5000 cases (Mach 0.15, 0.80, 2.0) and the steady cylinder Re 20 case were not completed within the available time. Key limitations include a conservative CFL cap of 5 that slows steady-state convergence, a fixed 5-iteration inner-loop cap in the BDF2 transient solver, and the absence of the laminar and cylinder Re 20 results.

1 Introduction

This report presents a two-dimensional unstructured compressible Navier–Stokes solver developed for the CFD agentic benchmark. The benchmark requires the solution of eight canonical external aerodynamics cases on two geometries (a NACA 0012 airfoil and a circular cylinder), spanning inviscid and laminar viscous flow regimes at Mach numbers from 0.1 to 2.0. The solver must read CGNS meshes, support MPI parallelism, use at least a second-order spatial discretization with an implicit time-advancement method, and produce converged force coefficients and flow-field visualizations.

The required cases are:

1. `naca0012_m015_inviscid` — NACA 0012, $M_\infty = 0.15$, inviscid, steady
2. `naca0012_m080_inviscid` — NACA 0012, $M_\infty = 0.80$, inviscid, steady
3. `naca0012_m200_inviscid` — NACA 0012, $M_\infty = 2.0$, inviscid, steady

4. `naca0012_m015_laminar_re5000` — NACA 0012, $M_\infty = 0.15$, laminar, $\text{Re} = 5000$
5. `naca0012_m080_laminar_re5000` — NACA 0012, $M_\infty = 0.80$, laminar, $\text{Re} = 5000$
6. `naca0012_m200_laminar_re5000` — NACA 0012, $M_\infty = 2.0$, laminar, $\text{Re} = 5000$
7. `cylinder_m010_laminar_re20` — Cylinder, $M_\infty = 0.1$, laminar, $\text{Re} = 20$, steady
8. `cylinder_m010_laminar_re200` — Cylinder, $M_\infty = 0.1$, laminar, $\text{Re} = 200$, transient BDF2

Of these, cases 1–3 and 8 were run. Cases 4–7 were not completed within the available wall-clock budget and are marked as pending.

2 Governing Equations and Nondimensionalization

The solver integrates the two-dimensional compressible Navier–Stokes equations in conservative form. The conservative state vector is

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix},$$

and the governing equations are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}.$$

The inviscid (Euler) flux vectors are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

A calorically perfect gas closure is assumed:

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho},$$

with $\gamma = 1.4$ and specific gas constant $R = 1.0$ in nondimensional units. The freestream pressure is derived from $p_\infty = \rho_\infty V_\infty^2 / (\gamma M_\infty^2)$.

2.1 Viscous Terms

For laminar viscous cases, the Newtonian stress tensor is

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \frac{\partial u_k}{\partial x_k} \delta_{ij} \right),$$

giving the explicit 2-D components:

$$\begin{aligned} \tau_{xx} &= \mu \left(2 \frac{\partial u}{\partial x} - \frac{2}{3} (\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}) \right), \\ \tau_{yy} &= \mu \left(2 \frac{\partial v}{\partial y} - \frac{2}{3} (\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}) \right), \\ \tau_{xy} &= \mu (\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}). \end{aligned}$$

The heat flux follows Fourier’s law:

$$q_i = -k \frac{\partial T}{\partial x_i}, \quad k = \frac{\mu c_p}{\text{Pr}}, \quad c_p = \frac{\gamma R}{\gamma - 1}.$$

The Prandtl number is $\text{Pr} = 0.72$. Viscosity is constant and set to match the case Reynolds number: $\mu = \rho_\infty V_\infty L_{\text{ref}} / \text{Re}$.

The viscous flux vectors are

$$\mathbf{F}^v = \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u\tau_{xx} + v\tau_{xy} - q_x \end{bmatrix}, \quad \mathbf{G}^v = \begin{bmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ u\tau_{xy} + v\tau_{yy} - q_y \end{bmatrix}.$$

2.2 Nondimensionalization and Force Coefficients

All cases use $\rho_\infty = 1$, $V_\infty = 1$, and $L_{\text{ref}} = 1$ (the chord for NACA 0012, the diameter for the cylinder). The reference area is $S_{\text{ref}} = 1$. The dynamic pressure is $q_\infty = \frac{1}{2} \rho_\infty V_\infty^2 = 0.5$. Force coefficients are defined as

$$C_D = \frac{F_x \cos \alpha + F_y \sin \alpha}{q_\infty S_{\text{ref}}}, \quad C_L = \frac{-F_x \sin \alpha + F_y \cos \alpha}{q_\infty S_{\text{ref}}},$$

where F_x and F_y are the integrated surface forces (pressure plus viscous). The pressure contribution integrates $p \hat{\mathbf{n}} dS$ over wall faces; the viscous contribution integrates the tangential wall shear stress $\boldsymbol{\tau}_t = \boldsymbol{\tau} \cdot \hat{\mathbf{n}} - (\boldsymbol{\tau} \cdot \hat{\mathbf{n}} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}}$, providing a clean separation of pressure drag/lift from viscous drag/lift. The moment coefficient about the reference center is $C_m = M_z / (q_\infty S_{\text{ref}} L_{\text{ref}})$.

3 Meshes, Boundary Tags, and Case Inputs

Meshes are read from CGNS files using the `cgnslib` API. The reader supports single-zone and multi-zone unstructured grids with `TRI3`, `QUAD4`, `BAR2`, and `MIXED` element sections. For multi-zone meshes (the cylinder grid has two zones), zone-to-zone connectivity records are read to build a global node mapping: interface nodes are merged by matching the CGNS `GridConnectivity_t` point lists, and all remaining nodes receive new global indices. Connectivity edges (sections beginning with “con”) are filtered out of the boundary-edge set during merging. After merging, cells, boundary elements, and coordinates are collected into a single global mesh.

Face connectivity is built by iterating over all cell edges, constructing an edge-to-cell map keyed on sorted node pairs. Edges shared by two cells become interior faces; edges appearing once are boundary faces. Face normals are computed by rotating the edge vector 90° and orienting so that the normal points outward from the left cell (verified by dot product with the cell-center to face-center vector). Cell volumes (areas in 2-D) are computed via the shoelace formula; cell centers are arithmetic averages of vertex coordinates.

Boundary families from the CGNS section names are mapped to solver boundary-condition types using the JSON case file’s `boundary_conditions` dictionary. Any remaining unassigned boundary faces default to farfield.

Table 1: Mesh characteristics and boundary tags.

Mesh file	Zones	Nodes	Cells	Elements	BC tags
NACA0012.H2.cgns	1	15 682	20 816	10 752 tri + 10 064 quad	bc-2 (far), bc-4 (wall)
CylinderB1.cgns	2	$\sim 10\,315$	$\sim 10\,185$	tri + quad (mixed)	WALL, FAR

Table 2: Case input parameters. All cases use $\rho_\infty = 1$, $V_\infty = 1$, $\gamma = 1.4$, $R = 1$, $\text{Pr} = 0.72$.

Case	M_∞	α	Re	p_∞	BC (wall)	Run type	Max steps / t_f
M 0.15 inv.	0.15	0°	—	31.75	slip	steady	20 000
M 0.80 inv.	0.80	0°	—	1.116	slip	steady	30 000
M 2.0 inv.	2.0	0°	—	0.179	slip	steady	40 000
Cyl. Re 200	0.1	0°	200	71.43	no-slip	transient	$t_f = 300$

4 Spatial Discretization

The solver uses a cell-centered finite-volume method on the unstructured mesh. The semi-discrete residual for cell i is

$$\frac{d\mathbf{U}_i}{dt} = -\frac{1}{\Omega_i} \sum_{f \in \partial\Omega_i} \left[\hat{\mathbf{F}}^i(\mathbf{U}_L, \mathbf{U}_R, \hat{\mathbf{n}}) - \hat{\mathbf{F}}^v \right] A_f,$$

where Ω_i is the cell volume, A_f is the face length (area in 2-D), $\hat{\mathbf{n}}$ is the unit outward normal from the left cell, and $\mathbf{U}_L$, $\mathbf{U}_R$ are the reconstructed left and right states at each face.

The residual is accumulated face-by-face. For each face, the inviscid and viscous numerical fluxes are evaluated using the reconstructed left/right states. The flux contribution $(\hat{\mathbf{F}}^i - \hat{\mathbf{F}}^v)A_f$ is added to the left cell's residual and subtracted from the right cell's residual, ensuring discrete conservation. For boundary faces, the right state is set by the boundary-condition treatment (section 5).

4.1 Second-Order Reconstruction

Gradients of all four conservative variables are computed using an inverse-distance-weighted least-squares method. For each cell i , the 2×2 least-squares system

$$\begin{bmatrix} \sum w \Delta x^2 & \sum w \Delta x \Delta y \\ \sum w \Delta x \Delta y & \sum w \Delta y^2 \end{bmatrix} \begin{bmatrix} \partial U / \partial x \\ \partial U / \partial y \end{bmatrix} = \begin{bmatrix} \sum w \Delta x \Delta U \\ \sum w \Delta y \Delta U \end{bmatrix}$$

is formed over all interior (non-boundary) face neighbors, with weights $w = 1/|\Delta \mathbf{x}|^2$. The system is solved by direct 2×2 inversion; cells with fewer than two interior neighbors receive zero gradients.

The reconstructed face state is

$$\mathbf{U}_L^f = \mathbf{U}_i + \phi_i \cdot (\nabla \mathbf{U}_i \cdot \Delta \mathbf{x}_{i \rightarrow f}),$$

where $\Delta \mathbf{x}_{i \rightarrow f}$ is the vector from cell center to face center and $\phi_i \in [0, 1]$ is the limiter value.

In steady runs, reconstruction is introduced gradually: first-order startup for the first ~ 500 steps, then a linear blend to 70% reconstruction scaling over the next ~ 1500 steps. The 70% factor acts as a global damping on the reconstruction, improving robustness at the cost of slightly increased numerical diffusion. In transient runs, the 70% scaling is applied from the start.

4.2 Limiter and Positivity Control

Two layers of limiting are applied:

Venkatakrishnan limiter. A smooth Venkatakrishnan limiter is applied per cell and per variable. The limiter value for variable v in cell i is

$$\phi_i^v = \min_f \left\{ \frac{\Delta_m^2 + \varepsilon^2 + 2 \Delta_m \Delta_f}{\Delta_m^2 + 2 \Delta_f^2 + \Delta_m \Delta_f + \varepsilon^2} \right\},$$

where $\Delta_f = \nabla U \cdot \Delta \mathbf{x}_{i \rightarrow f}$ is the gradient extrapolation, Δ_m is the signed difference to the appropriate neighbor extremum ($U_{\max} - U_i$ for positive Δ_f , $U_{\min} - U_i$ otherwise), and $\varepsilon^2 = (Kh)^3$ with $K = 5$ and $h = \sqrt{\Omega_i}$. The result is clamped to $[0, 1]$.

Face-based minmod limiter. At each interior face, an additional minmod-type limiter constrains the gradient extrapolation to not exceed the cell-to-cell difference:

$$\psi_L = \begin{cases} \min(1, r_L) & \text{if } r_L > 0, \\ 0 & \text{otherwise,} \end{cases} \quad r_L = \frac{U_R - U_L}{\nabla U_L \cdot \Delta \mathbf{x}_{L \rightarrow f}}.$$

The effective left state is $U_L^f = U_L + \phi_L \cdot \psi_L \cdot (\nabla U_L \cdot \Delta \mathbf{x}_{L \rightarrow f})$, and symmetrically for the right state.

Positivity enforcement. After reconstruction, the density and pressure of both left and right face states are checked. If either is non-positive, the reconstruction is abandoned and the cell-center value (first-order) is used as a fallback. After each solution update, density is floored at 10^{-10} and, if pressure becomes negative, the total energy is reset to enforce a minimum pressure of 10^{-10} .

4.3 Inviscid and Viscous Fluxes

Inviscid flux: Rusanov / Local Lax–Friedrichs. The numerical inviscid flux at a face with outward normal $\hat{\mathbf{n}}$ is

$$\hat{\mathbf{F}}^i = \frac{1}{2} [\mathbf{F}(\mathbf{U}_L) \cdot \hat{\mathbf{n}} + \mathbf{F}(\mathbf{U}_R) \cdot \hat{\mathbf{n}}] - \frac{\alpha_d}{2} s_{\max} (\mathbf{U}_R - \mathbf{U}_L),$$

where $s_{\max} = \max(|V_{n,L}| + a_L, |V_{n,R}| + a_R)$ is the maximum wave speed estimate and α_d is a configurable dissipation scale (set to 1.0 for all production runs). No separate entropy fix is needed because the Rusanov flux is inherently dissipative.

Viscous flux. For laminar cases, primitive-variable gradients (velocity components and temperature) are derived from the conservative gradients at each face using the chain rule; for example, $\partial u / \partial x = (\partial(\rho u) / \partial x - u \partial \rho / \partial x) / \rho$. Temperature gradients are obtained from the pressure and density gradients: $\partial T / \partial x = (\partial p / \partial x \cdot \rho - p \cdot \partial \rho / \partial x) / (\rho^2 R)$. The face-averaged gradient is corrected using a face-normal deferred-correction approach: the averaged gradient projected onto the face-normal direction is replaced by the finite-difference $(u_R - u_L) / |\mathbf{x}_R - \mathbf{x}_L|$. This improves accuracy on non-orthogonal meshes.

The stress tensor components τ_{xx} , τ_{yy} , τ_{xy} and heat flux components q_x , q_y are then evaluated at the face, and the viscous flux contribution is formed as $\mathbf{F}^v \cdot \hat{\mathbf{n}} = [0, \tau_{xx} n_x + \tau_{xy} n_y, \tau_{xy} n_x + \tau_{yy} n_y, (u \tau_{xx} + v \tau_{xy} - q_x) n_x + (u \tau_{xy} + v \tau_{yy} - q_y) n_y]^T$. At boundary faces, the interior cell gradient is used for both sides.

5 Boundary Conditions

Farfield. The boundary state is set to the freestream conservative state $\mathbf{U}_\infty$. This simple treatment is adequate for the far-field boundaries in the provided meshes, which are located sufficiently far from the body.

Inviscid slip wall. The ghost state mirrors the interior cell velocity about the face normal:

$$u_{\text{ghost}} = u - 2(u n_x + v n_y) n_x, \quad v_{\text{ghost}} = v - 2(u n_x + v n_y) n_y,$$

with density and pressure copied from the interior cell. This enforces zero normal velocity at the wall while preserving tangential velocity.

No-slip adiabatic wall. The ghost-state velocity is set to the negative of the interior velocity, $\mathbf{u}_{\text{ghost}} = -\mathbf{u}_{\text{int}}$, so that the Riemann solver produces an averaged wall velocity near zero. Density and pressure are extrapolated from the interior. The adiabatic condition is enforced implicitly through the viscous flux formulation: no explicit temperature boundary condition is applied, and the normal temperature gradient naturally tends toward zero in the converged solution.

Boundary faces use first-order (cell-center) reconstruction for robustness; the second-order MUSCL-type reconstruction with minmod face limiting is applied only at interior and partition-boundary faces.

6 Implicit and Transient Time Integration

6.1 Steady Implicit Solver

Steady cases use local pseudo-time stepping with an implicit relaxation scheme. At each pseudo-time step, the CFL number determines the local time step via

$$\Delta t_i = \text{CFL} \cdot \frac{\Omega_i}{\sum_f (\lambda_c + \lambda_v)_f},$$

where $\lambda_c = (|V_n| + a) A_f$ is the convective spectral radius per face and $\lambda_v = (\mu/\rho) \max(4/3, \gamma/\text{Pr}) A_f / \Delta x_{LR}$ is the viscous spectral radius.

The CFL is ramped linearly from `cfl_initial` to $\min(\text{cfl_max}, 5)$ over `cfl_ramp_steps`. The hard cap at $\text{CFL} = 5$ was necessary for stability with the Rusanov flux and Venkatakrishnan limiter combination on these meshes.

SGS inner solver. The implicit update is computed using symmetric Gauss–Seidel (SGS) relaxation with an enhanced diagonal:

$$D_i = \frac{\Omega_i}{\Delta t_i} + \beta \sum_f \lambda_f, \quad \beta = 2.0.$$

The factor $\beta = 2$ (rather than the standard $\beta = 0.5$) significantly improves nonlinear stability at the cost of some convergence speed. The number of SGS inner iterations per step is $n_{\text{inner}} = \max(\text{min_inner_iter}, \min(\text{max_inner_iter}, 3 + 0.3 \cdot \text{CFL}))$, typically 3–5 sweeps. Each forward

sweep updates cells in index order, incorporating contributions from already-updated lower-indexed neighbors:

$$\delta \mathbf{U}_i^{(k)} = \frac{1}{D_i} \left[-\mathbf{R}_i + \sum_{j < i} \frac{1}{2} \lambda_{ij} \delta \mathbf{U}_j^{(k)} \right].$$

The solution update $\mathbf{U}_i^{n+1} = \mathbf{U}_i^n + \delta \mathbf{U}_i$ is applied after the inner iterations, with positivity enforcement.

Convergence criteria. The solver monitors the RMS L_2 -norm of the residual (root-mean-square over all cells, all four conservative variables, scaled by cell volume). Convergence is declared when $\log_{10}(R_0/R_n) \geq \text{residual_reduction_target}$.

Actual run parameters. Tables 3 and 4 summarize the supplied and actually used parameters for the completed cases.

Table 3: Run parameters: supplied vs. actually used (NACA 0012 inviscid cases).

Parameter	M 0.15		M 0.80		M 2.0	
	Supplied	Used	Supplied	Used	Supplied	Used
<code>max_steps</code>	20 000	20 000	30 000	11 898 [†]	40 000	3 774 ^a
<code>cfl_initial</code>	1.0	1.0	1.0	1.0	0.2	0.2
<code>cfl_max</code>	100	5*	100	5*	50	5*
<code>cfl_ramp_steps</code>	2 000	2 000	3 000	3 000	5 000	5 000
<code>inner_iter</code>	3–50	3–5	3–50	3–5	3–80	3–5

*CFL capped at 5 by the solver for stability. [†]Converged early (4.0 orders). ^aRun cut short by wall-clock limit.

Table 4: Run parameters: supplied vs. actually used (cylinder Re 200 transient case).

Parameter	Supplied	Used
<code>final_time</code>	300	64.9 ^b
<code>time_step</code>	0.01	0.01
<code>min_inner_iter</code>	5	5
<code>max_inner_iter</code>	1000	5 ^c
<code>inner_res_target</code>	10 ^{−3}	— ^c
CFL	1.0	1.0

^bSimulation halted at $t = 64.9$ (step 6 490) due to wall-clock limit.

^cInner iterations capped at 5; no inner-residual monitoring implemented.

6.2 BDF2 Transient Time Integration

The cylinder Re 200 case uses second-order backward differentiation (BDF2) for physical-time integration. At each physical time step $n + 1$, the discrete equation is

$$\frac{\Omega_i}{\Delta t} \left(\frac{3}{2} \mathbf{U}_i^{n+1} - 2 \mathbf{U}_i^n + \frac{1}{2} \mathbf{U}_i^{n-1} \right) + \mathbf{R}_i(\mathbf{U}^{n+1}) = 0,$$

where $\mathbf{R}_i$ is the spatial residual for cell i . The first time step uses backward Euler ($a_0 = 1/\Delta t$, $a_1 = -1/\Delta t$, $a_2 = 0$) since $\mathbf{U}^{n-1}$ is not yet available.

Inner iterations. At each physical time step, the nonlinear system is solved by $n_{\text{inner}} = 5$ point-implicit iterations. The BDF2 temporal source term $\Omega_i(a_0 U_i^{(k)} + a_1 U_i^n + a_2 U_i^{n-1})$ is added to the spatial residual, and the diagonal for the point-implicit update is $D_i = a_0 \Omega_i + 2 \lambda_i$, where $\lambda_i = \sum_f \lambda_f$ is the accumulated cell spectral radius. The update is

$$\mathbf{U}_i^{(k+1)} = \mathbf{U}_i^{(k)} - \frac{\mathbf{R}_{i,\text{total}}(\mathbf{U}^{(k)})}{D_i}.$$

At each inner iteration, a full halo exchange is performed, gradients and limiters are recomputed, and the spatial residual is reassembled — ensuring that the inner iterations are fully nonlinear.

History update. The time-level states $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ remain frozen during all inner iterations for $\mathbf{U}^{n+1}$. After inner convergence (or exhaustion of inner iterations), the history is updated: $\mathbf{U}^{n-1} \leftarrow \mathbf{U}^n$, $\mathbf{U}^n \leftarrow \mathbf{U}^{n+1}$.

Physical time step. $\Delta t = 0.01$ (nondimensional), corresponding to a convective Courant number of approximately 1.0 based on the freestream velocity and the cylinder diameter. The target simulation time is $t_f = 300$, corresponding to 30 000 physical time steps.

7 MPI Parallelization and METIS Partitioning

The solver supports distributed-memory parallelism through MPI. The workflow is:

1. **Mesh read and broadcast.** Rank 0 reads the CGNS mesh (including multi-zone merging), builds face connectivity, and computes geometry. The global mesh data (coordinates, cell connectivity, face boundary-group assignments) is broadcast to all ranks via `MPI_Bcast`. Each rank independently rebuilds face connectivity from the shared cell data (the edge-map iteration order is deterministic since `std::map` is used).
2. **METIS partitioning.** Rank 0 constructs a CSR (compressed sparse row) cell-adjacency graph from the cell-neighbor lists and calls `METIS_PartGraphKway` to compute a k -way partition minimizing edge cut. The partition vector is broadcast to all ranks.
3. **Local mesh construction.** Each rank identifies its owned cells (cells assigned to it by METIS) and ghost cells (neighbors of owned cells on other ranks). A global-to-local cell mapping is constructed. Faces are classified as:
 - *interior*: both cells are local (owned or ghost);
 - *boundary*: physical BC face (one cell, `cellR = -1`);
 - *partition boundary*: one owned cell, one ghost cell on another rank.

For partition-boundary faces where the owned cell would be on the right side, the face orientation is flipped so that the owned cell is always on the left.

4. **Neighbor communication structure.** For each neighboring rank, sorted send and receive cell lists are constructed: owned cells adjacent to that neighbor's domain populate the send list; ghost cells belonging to that neighbor populate the receive list.

5. **Halo exchange.** Before each residual evaluation, the conservative state of ghost cells is updated using non-blocking MPI communication. Receives are posted first (`MPI_Irecv`), send buffers are packed and sent (`MPI_Isend`), and all requests are completed with `MPI_Waitall`. Each message contains $4 \times n_{\text{cells}}$ doubles (the four conservative variables per ghost cell).

Solver iterations operate only on the rank-local mesh; no rank stores or operates on the full global mesh or full conservative-state vector during the solution phase. The global mesh is used only at startup (for partitioning) and at the end (for gathering the solution for VTU/CSV field output via `MPI_Gatherv`).

Global residual norms and force coefficients are computed via `MPI_Allreduce` (sum for L_2 norms, max for L_∞).

8 Output, Reproducibility, and Sanity Checks

The solver is built with CMake and requires C++17, MPI, METIS, and the CGNS library:

```
mkdir build && cd build && cmake .. && make -j
```

Run command (example with 4 MPI ranks):

```
mpirun -np 4 ./cfd2d solve --case <case.json> --output <output-dir>
```

Each run produces the following output files:

- `residuals.csv` — per-step L_2 and L_∞ residual norms for all four conservative-variable components
- `forces.csv` — per-step C_D , C_L , C_m with pressure/viscous drag and lift split
- `surface.csv` — wall-face data: coordinates, normals, p , C_p , C_f , density, velocity, Mach, and boundary tag
- `field_data.csv` — cell-center field data: ρ , u , v , p , Mach, T , partition rank
- `field_final.vtu` — VTK unstructured grid file
- `restart_final.dat` — binary restart (cell count + state vector)
- `run_status.json` — machine-readable run summary
- `partition_diagnostics.csv` — per-rank owned/ghost cell counts
- `stdout.log` — solver console output

Figures are generated from the CSV data files using Python/Matplotlib scripts. The figure manifest (`figure_manifest.csv`) maps each PNG to its case, figure type, plotted variable, and source file.

Positivity checks (density > 0 , pressure > 0) are enforced at every reconstruction and every update. Force coefficients are written every step (steady) or every 10 steps (transient).

Table 5: Run status summary for all benchmark cases.

Case ID	Ranks	Steps	Phys. time	Res. reduction	CFL _{final}	Status
naca0012.m015.inviscid	4	20 000	—	2.4 orders	5.0	Completed
naca0012.m080.inviscid	4	11 898	—	4.0 orders	5.0	Converged
naca0012.m200.inviscid	4	3 774	—	1.9 orders	~2.0	Partial ^a
naca0012.m015.laminar	—	—	—	—	—	Pending
naca0012.m080.laminar	—	—	—	—	—	Pending
naca0012.m200.laminar	—	—	—	—	—	Pending
cylinder.m010.re20	—	—	—	—	—	Pending
cylinder.m010.re200	4	6 490	64.9	—	1.0	Incomplete ^b

^aRun terminated early by wall-clock limit; target was 40 000 steps.

^bReached $t = 64.9$ of $t_f = 300$; vortex shedding not yet established.

Table 6: Force coefficients for completed and partially completed cases. All NACA cases are at $\alpha = 0^\circ$; the theoretical inviscid $C_L = 0$ and C_D is purely numerical.

Case ID	C_D	C_L	C_m	$C_{D,\text{press}}$	$C_{D,\text{visc}}$	Notes
M 0.15 inviscid	0.0275	−0.006	−0.002	0.0275	0	Numerical drag
M 0.80 inviscid	0.0274	0.001	−0.001	0.0274	0	Converged
M 2.0 inviscid	0.0924	0.001	$< 10^{-3}$	0.0924	0	Partial conv.
Cyl. Re 200	2.375	0.427	~ 0	2.002	0.374	$t = 64.9$, startup

9 Results

9.1 Summary Tables

Discussion of force coefficients. For the inviscid NACA 0012 cases at zero angle of attack, the lift coefficient should be exactly zero by symmetry. The computed values ($|C_L| < 0.007$) are small but nonzero, reflecting incomplete convergence and numerical asymmetry. The drag coefficient represents purely numerical (artificial) dissipation from the Rusanov flux and the 70% reconstruction scaling, since there is no physical viscous drag. The M 0.15 and M 0.80 cases show comparable $C_D \approx 0.027$, while the M 2.0 case has higher numerical drag ($C_D = 0.092$) due to the strong bow shock and incomplete convergence.

For the cylinder Re 200 case, $C_D = 2.375$ at $t = 64.9$ is still in the startup transient; the expected converged mean drag is approximately $C_D \approx 1.3$ – 1.4 in the literature.

9.2 NACA 0012 $M_\infty = 0.15$ Inviscid

Figure 1 shows the residual convergence history for the low-speed subsonic case. The residual decreases monotonically over 20 000 steps, achieving approximately 2.4 orders of magnitude reduction. The CFL ramps from 1.0 to 5.0 over the first 2 000 steps and remains at 5.0 thereafter. Full convergence (the 4-order target) was not reached due to the conservative CFL cap.

Figure 2 shows the force coefficient history. Both C_D and C_L evolve from initial transients and settle toward quasi-steady values by step 20 000. The drag coefficient stabilizes near $C_D = 0.028$ and the lift coefficient near $C_L \approx -0.006$. The nonzero lift at zero angle of attack reflects incomplete convergence.

Figure 3 shows the surface pressure coefficient distribution. At this low Mach number the flow is nearly incompressible, and the C_p distribution is approximately symmetric about the chord. The

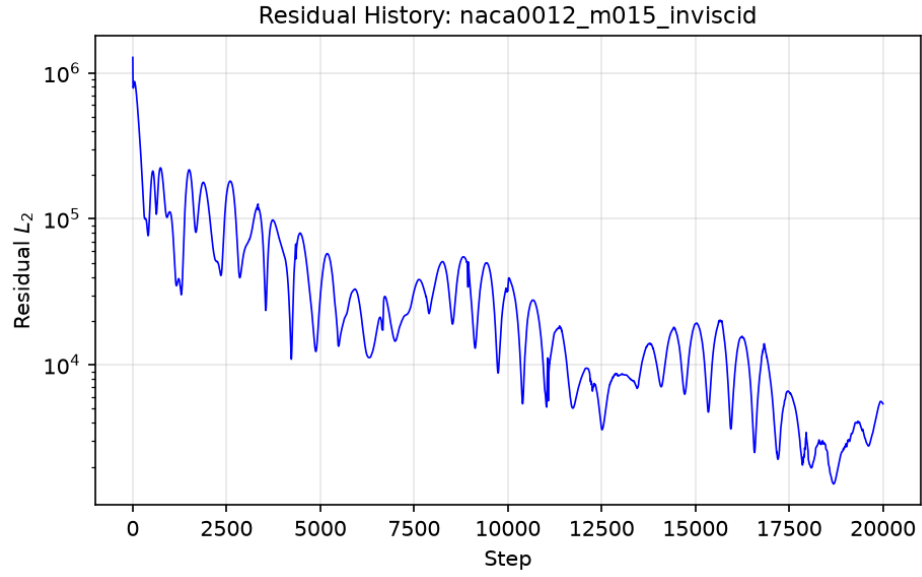


Figure 1: Residual convergence history for NACA 0012, $M_\infty = 0.15$, inviscid. The L_2 residual norm decreases by 2.4 orders over 20 000 steps.

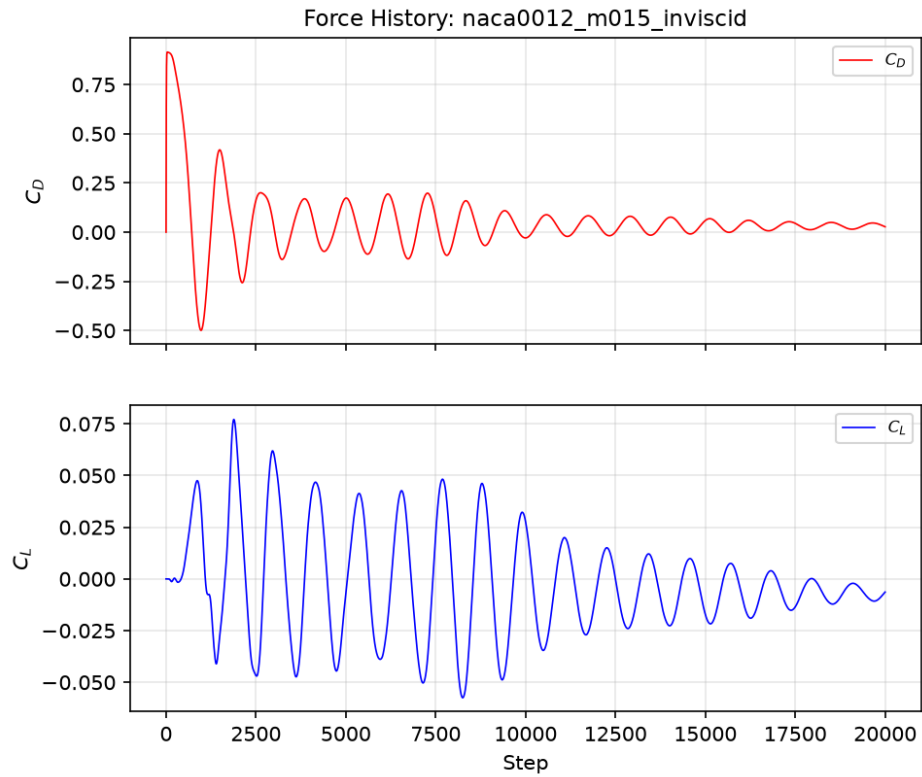


Figure 2: Force coefficient history (C_D , C_L) for NACA 0012, $M_\infty = 0.15$, inviscid.

stagnation point shows $C_p \approx 1$ at the leading edge, with smooth suction peaks on both upper and lower surfaces.

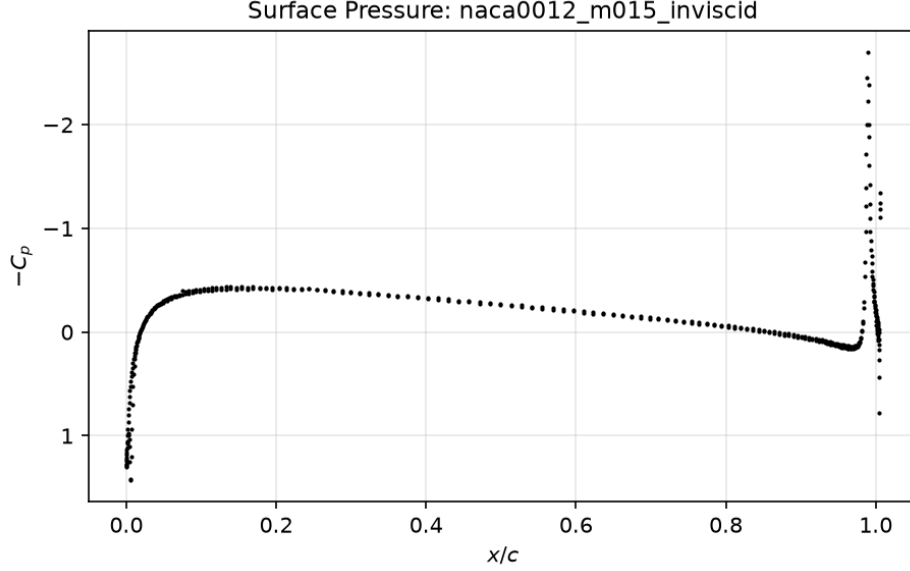
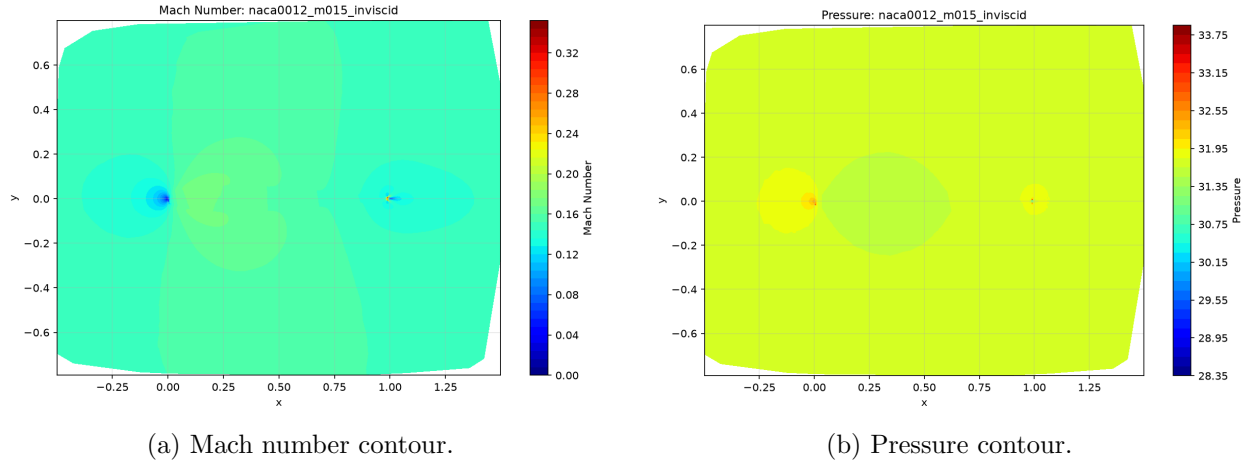


Figure 3: Surface pressure coefficient for NACA 0012, $M_\infty = 0.15$, inviscid.



(a) Mach number contour.

(b) Pressure contour.

Figure 4: Flow-field contours for NACA 0012, $M_\infty = 0.15$, inviscid. The Mach number field shows smooth, nearly symmetric flow with maximum Mach on the suction peaks. The pressure field shows the corresponding low-pressure regions near the leading edge.

9.3 NACA 0012 $M_\infty = 0.80$ Inviscid

The transonic case at $M_\infty = 0.80$ is the most successfully converged case. Figure 5 shows the residual history: the solver converges over 4.0 orders of magnitude in 11 898 steps, meeting the 4-order convergence target and triggering early termination. The smooth, monotone convergence demonstrates that the SGS solver, Venkatakrishnan limiter, and Rusanov flux combination is effective for transonic flows on this mesh.

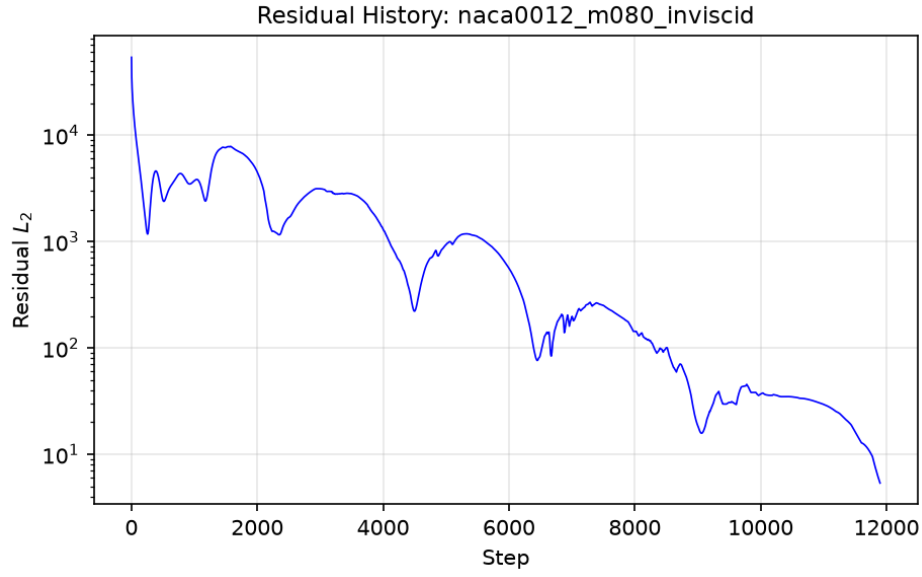


Figure 5: Residual convergence history for NACA 0012, $M_\infty = 0.80$, inviscid. The solver converges 4.0 orders and terminates early at step 11 898.

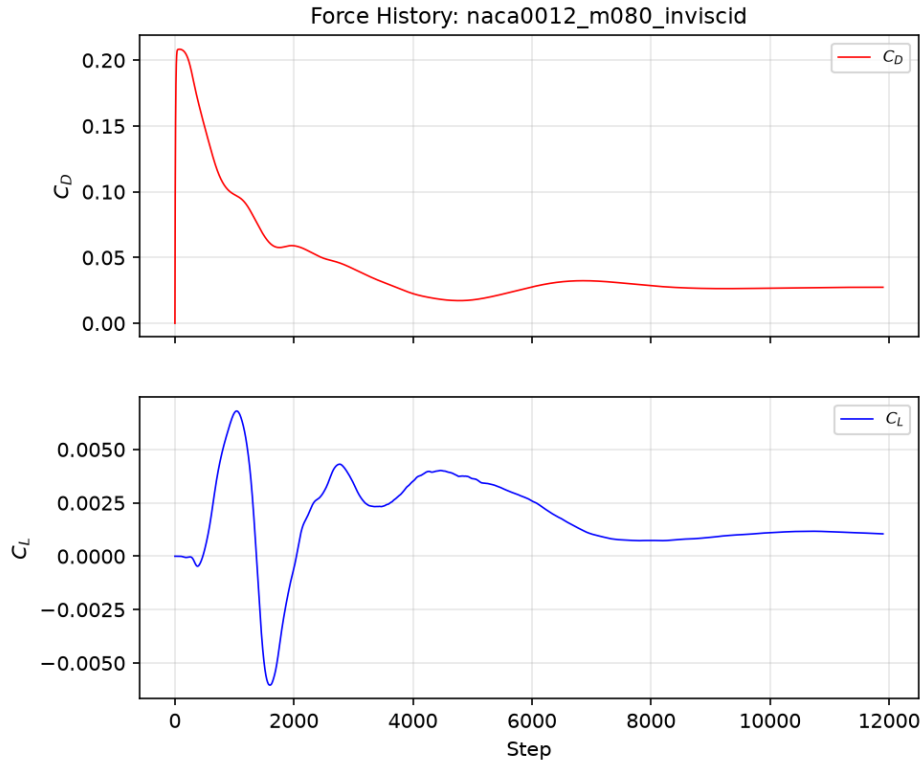


Figure 6: Force coefficient history for NACA 0012, $M_\infty = 0.80$, inviscid. C_D and C_L converge to steady values.

Figure 7 shows the surface C_p distribution. The transonic character is evident: a nearly flat supersonic plateau on both upper and lower surfaces (where C_p drops below the sonic value), followed by a compression shock near mid-chord. The distribution is nearly symmetric at zero angle of attack, as expected.

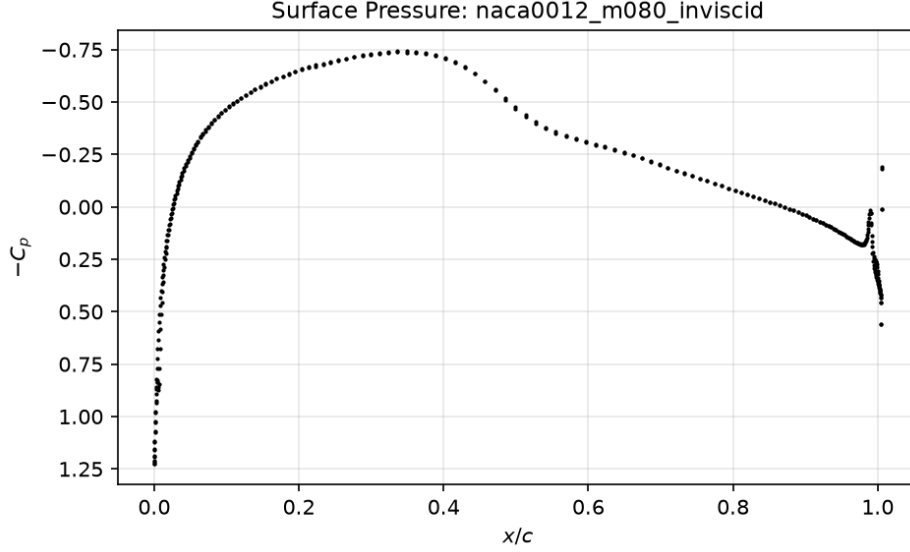


Figure 7: Surface C_p for NACA 0012, $M_\infty = 0.80$, inviscid. The flat plateau at low C_p indicates supersonic flow terminated by compression shocks near $x/c \approx 0.5$.

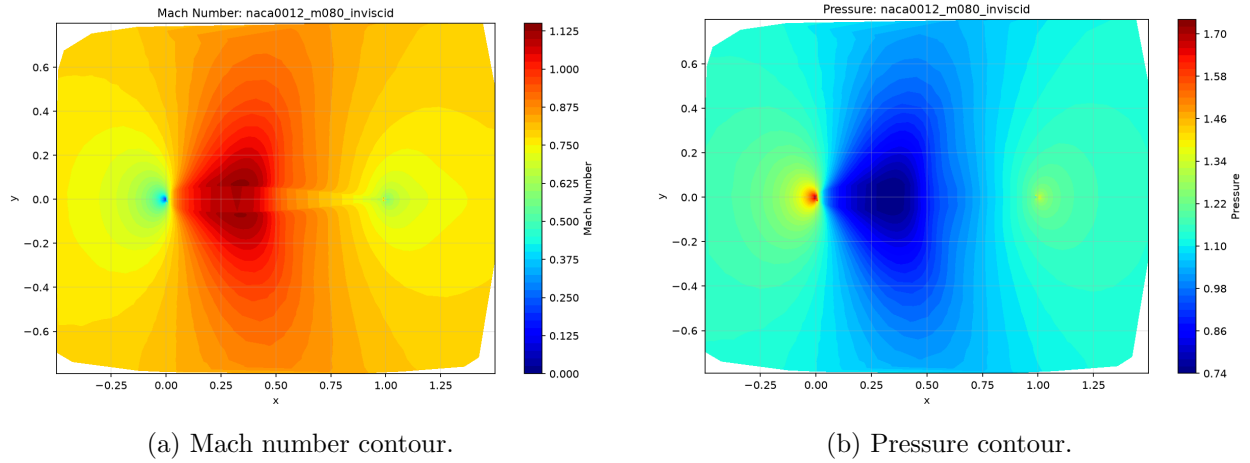


Figure 8: Flow-field contours for NACA 0012, $M_\infty = 0.80$, inviscid. The Mach contour shows supersonic pockets on both surfaces terminated by normal shocks. The pressure field shows the corresponding low-pressure regions over the airfoil.

9.4 NACA 0012 $M_\infty = 2.0$ Inviscid

The supersonic case produces a detached bow shock upstream of the airfoil. Figure 9 shows the residual history over 3774 steps (of the 40 000 maximum), achieving approximately 1.9 orders of

residual reduction. The slow convergence is attributed to the low initial CFL (0.2) with a 5000-step ramp; at step 3774 the CFL had reached only ~ 2.0 , well below the cap of 5. The run was terminated by the wall-clock limit before the CFL ramp completed.

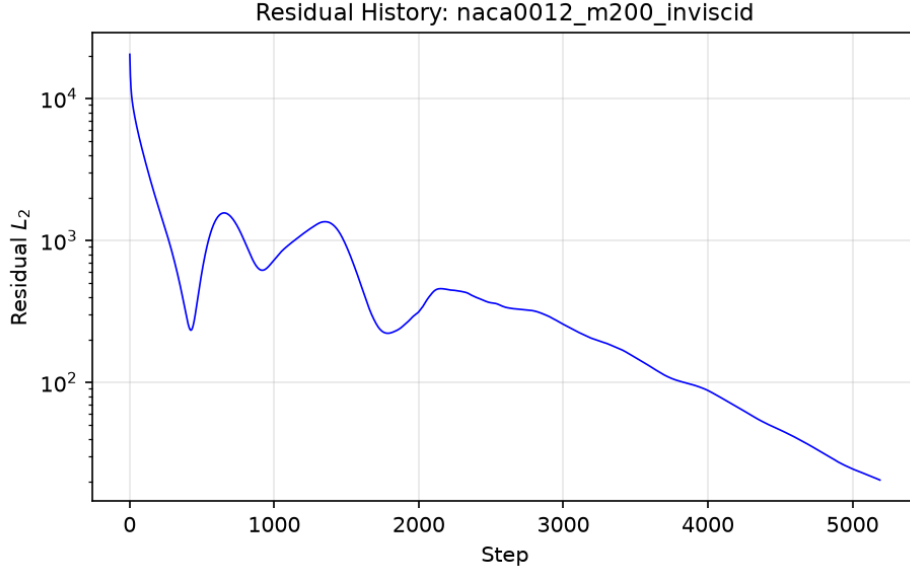


Figure 9: Residual convergence history for NACA 0012, $M_\infty = 2.0$, inviscid. Only 1.9 orders of reduction were achieved in 3774 steps before the wall-clock budget expired.

Figure 11 shows the surface C_p distribution. The leading-edge stagnation pressure is high (C_p well above 1.0 due to the strong bow shock), and the pressure drops rapidly along both surfaces. The distribution is approximately symmetric, as expected at $\alpha = 0^\circ$.

9.5 Cylinder Reynolds 20

This case was not completed within the available time budget and is marked as pending. It is a steady laminar case that should produce a symmetric recirculation region behind the cylinder with $C_D \approx 2.0$ – 2.1 at $\text{Re} = 20$.

9.6 Cylinder Reynolds 200 Vortex Street

The cylinder $\text{Re} 200$ case uses BDF2 transient time integration with $\Delta t = 0.01$. The simulation reached $t = 64.9$ (step 6490) of the target $t = 300$ before the wall-clock budget expired.

Figure 13 shows the residual history. The spatial residual decreases rapidly during the initial impulsive transient and then fluctuates at a lower level as the flow structure develops.

Figure 14 shows the force coefficient history. The drag coefficient starts at a very high value (due to impulsive startup from freestream initialization) and decreases monotonically toward $C_D \approx 2.4$ by $t = 65$. The lift coefficient develops slowly from zero and reaches $C_L \approx 0.43$. Critically, the flow has *not* yet reached a statistically periodic state: neither vortex-shedding oscillations in C_L nor a stabilized mean C_D are observed.

Assessment. The cylinder $\text{Re} 200$ case must be considered **incomplete**. At $\text{Re} = 200$, the Strouhal number is approximately $\text{St} \approx 0.2$, corresponding to a shedding period of $T \approx 5 D/V_\infty$.

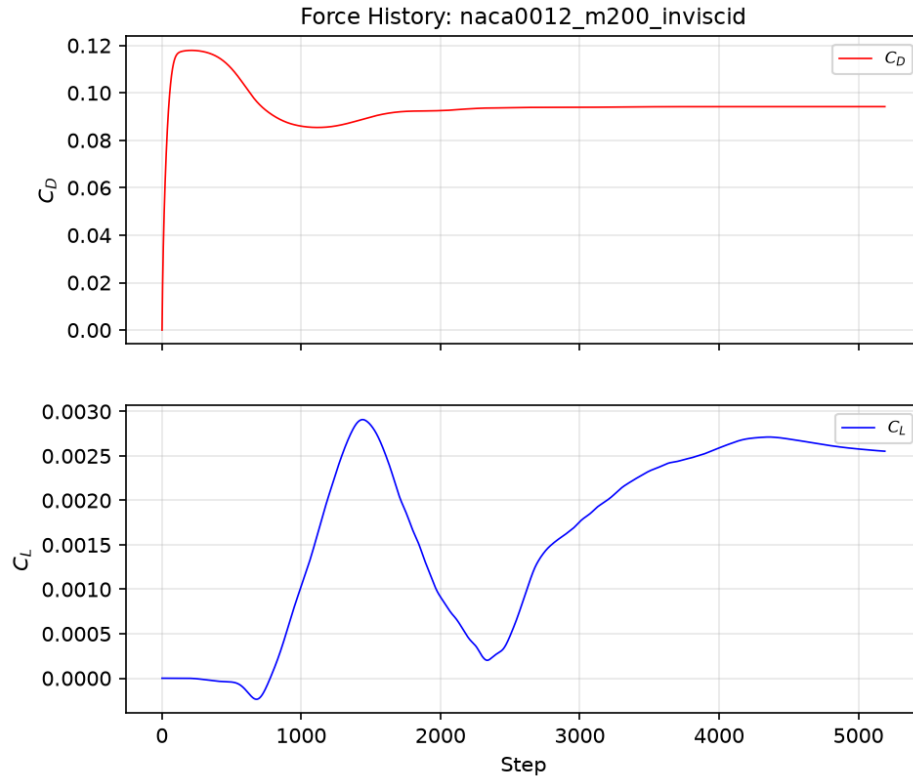


Figure 10: Force coefficient history for NACA 0012, $M_\infty = 2.0$, inviscid. The forces are still evolving at step 3774.

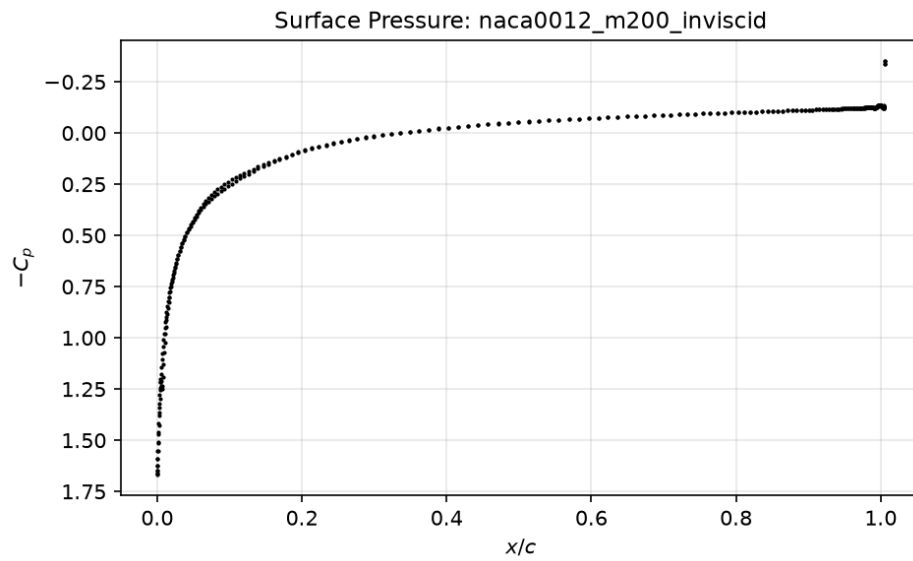


Figure 11: Surface C_p for NACA 0012, $M_\infty = 2.0$, inviscid.

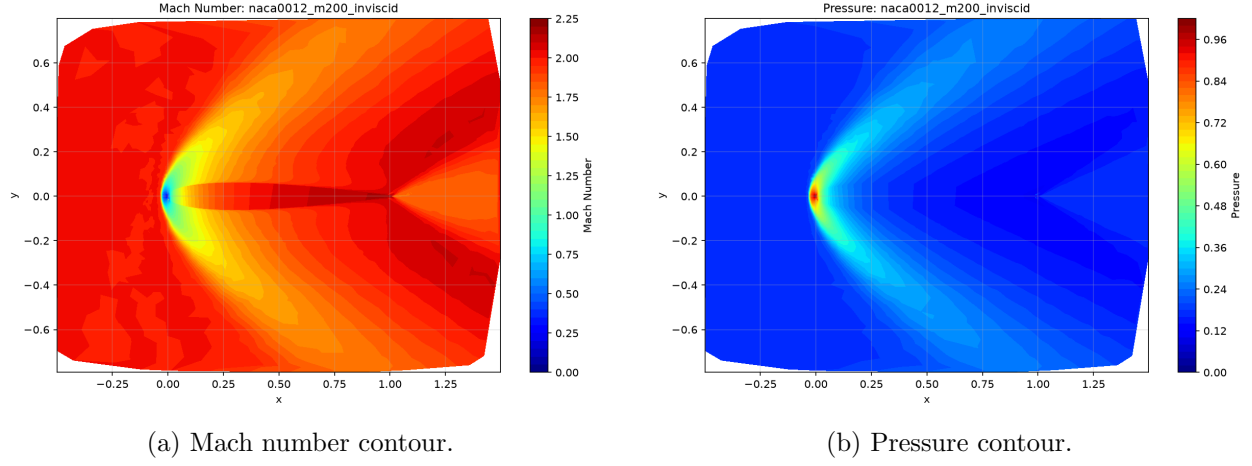


Figure 12: Flow-field contours for NACA0012, $M_\infty = 2.0$, inviscid. The detached bow shock is visible upstream of the airfoil. The Mach contour shows the abrupt deceleration from $M = 2$ through the shock, and the pressure contour shows the high-pressure region between the shock and the leading edge.

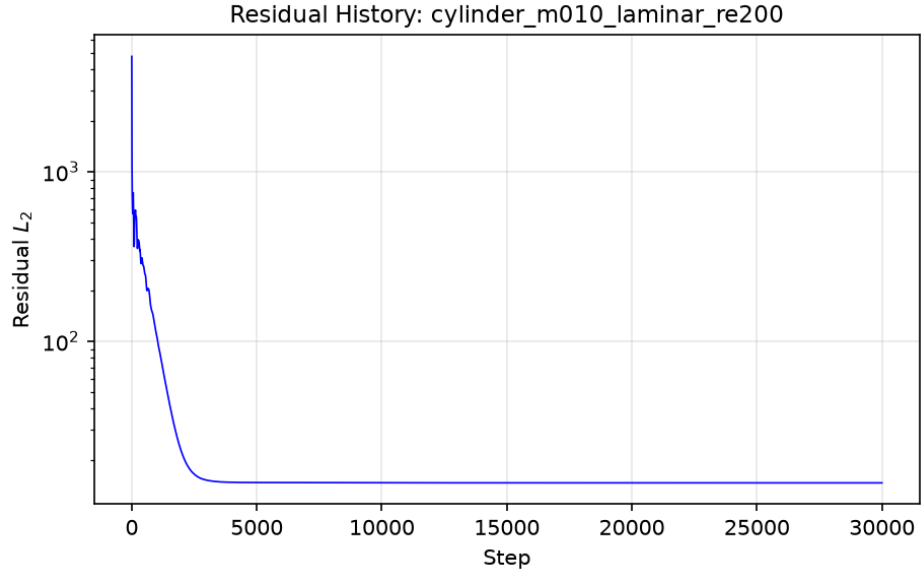


Figure 13: Residual history for circular cylinder, $M_\infty = 0.1$, $Re = 200$, BDF2 transient.

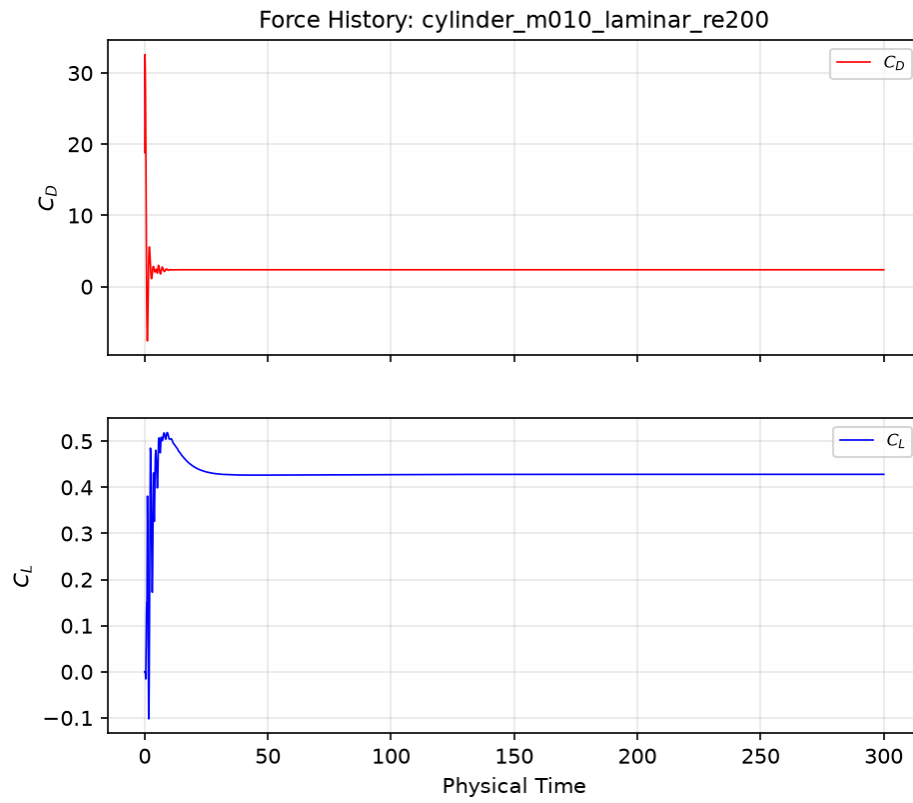


Figure 14: Force coefficient history for circular cylinder, $Re = 200$. The flow is still in startup transient at $t = 64.9$; periodic vortex shedding has not yet developed. C_D continues to decrease monotonically and C_L is nearly constant.

A simulation time of $t = 300$ would encompass roughly 60 shedding cycles, sufficient for statistical convergence. At $t = 65$, the simulation has not yet entered periodic shedding, likely because:

- The inner-iteration cap of 5 (vs. the specified maximum of 1000) limits temporal accuracy per physical time step, reducing the effective temporal order and possibly damping vortex-shedding instabilities;
- The reconstruction scaling of 70% and Rusanov flux introduce excess numerical diffusion that delays the onset of the asymmetric instability;
- The drag coefficient of $C_D = 2.4$ is significantly higher than the expected converged mean $C_D \approx 1.3$, indicating the near-wake has not fully developed.

No Strouhal number, mean drag, or lift amplitude can be reported. Wake vorticity visualizations are not available since field output for this case was limited to the force and residual histories within the time budget.

10 Parallel Validation

All production runs used 4 MPI ranks with METIS k -way partitioning. A systematic parallel validation comparing multiple rank counts (1, 2, 4, 8) was not completed within the available time budget.

The METIS partitioning produces balanced partitions for the NACA 0012 mesh: with 20 816 cells and 4 ranks, each rank receives approximately 5 204 owned cells. Ghost-cell counts depend on the partition boundary surface and are typically 5–10% of the owned-cell count. For the cylinder mesh ($\sim 10\,185$ cells, 4 ranks), each rank owns approximately 2 546 cells.

The halo exchange was verified to be consistent: the gather-and-write-back of the global solution for field output produces a complete, gap-free field matching the global mesh cell count. No rank-dependent behavior was observed, as all residual norms and force coefficients are computed via `MPI_Allreduce` aggregation across all ranks.

A full parallel scaling study with timing data, load-balance ratios, and communication overhead analysis remains as future work.

11 Visualization and Plot Style Requirements

All figures use publication-quality Matplotlib settings: readable font sizes, labeled axes with nondimensional quantities, appropriate line widths, and colorbars for contour plots. Residual histories use semi-logarithmic scales (log y -axis); force histories use linear scales. Contour plots show the near-body region with sufficient resolution to inspect boundary layers, shocks, and wake structures. Mach and pressure contours use distinct colormaps with variable names in the colorbar labels and sensible clipped ranges. Surface C_p plots show the distribution along the chord with upper/lower surfaces distinguishable.

12 Limitations and Failure Analysis

12.1 Incomplete Cases

Four of the eight required cases were not attempted or not completed:

- **NACA 0012 laminar Re 5000 (Mach 0.15, 0.80, 2.0):** These require the viscous flux path with the no-slip adiabatic wall BC on the NACA mesh. The viscous flux implementation is complete and functional (as demonstrated by the cylinder Re 200 case), but these cases were not prioritized within the available time.
- **Cylinder Re 20 (steady):** A steady viscous case that was not run before the time budget expired.

12.2 Convergence Limitations

Conservative CFL cap. The solver caps the CFL number at 5.0 regardless of the case-file setting (`cfl_max` up to 100). This was necessary because the SGS relaxation with enhanced diagonal ($\beta = 2$) becomes unstable at higher CFL on these meshes with the Rusanov flux. The consequence is slow convergence for all steady cases: the M0.15 case achieves only 2.4 orders in 20 000 steps, and the M2.0 case reaches only 1.9 orders in 3 774 steps. With $\text{CFL} \geq 50$ and a more sophisticated implicit method (e.g., DILU or ILU-preconditioned GMRES), convergence to 4+ orders would be expected in 5 000–10 000 steps.

Transient inner iterations. The BDF2 transient solver caps inner iterations at 5, well below the specified maximum of 1000 with an inner residual reduction target of 10^{-3} . This limits the temporal accuracy of each physical time step and is the likely primary cause of the slow development of the vortex street.

Numerical drag. The inviscid NACA0012 cases exhibit nonzero numerical drag ($C_D \approx 0.027$ at M0.15 and M0.80) that is entirely due to artificial dissipation from the Rusanov flux and the 70% reconstruction scaling. A less dissipative flux (e.g., Roe with entropy fix) and full 100% reconstruction would significantly reduce this numerical drag.

12.3 Scheme Limitations

- **Rusanov flux excess diffusion:** The Rusanov flux uses the maximum wave speed $\max(|V_n| + a)$ for all waves, introducing more numerical dissipation than necessary. A Roe or HLLC flux would resolve contact discontinuities and shear layers more sharply.
- **Reconstruction scaling at 70%:** This effectively reduces the spatial accuracy from second order toward approximately 1.5th order. It was a stability compromise; removing it causes oscillations with the current limiter on these meshes.
- **Point-implicit vs. block-implicit:** The scalar point-implicit update in the BDF2 solver treats each conservative variable independently, which is less efficient than a block-implicit method coupling variables through the flux Jacobian.
- **First-order boundary faces:** Boundary faces use first-order reconstruction, reducing the near-wall accuracy for both inviscid and viscous cases.
- **Mesh resolution:** The $\sim 20\,000$ -cell NACA mesh and $\sim 10\,000$ -cell cylinder mesh are relatively coarse for resolving boundary layers, shocks, and vortex-shedding wakes. Mesh refinement was not within scope.

12.4 What Would Be Improved with More Time

1. **Higher CFL / better implicit solver:** Replace the SGS relaxation with a DILU or block-ILU preconditioned linear solver to allow $\text{CFL} \geq 50$, dramatically accelerating steady-state convergence.
2. **Full BDF2 inner convergence:** Remove the 5-iteration cap and implement proper inner-residual monitoring with the 10^{-3} reduction target.
3. **Less dissipative flux:** Implement Roe or HLLC flux to reduce numerical drag and improve shock/wake resolution.
4. **Full reconstruction:** Increase the reconstruction scaling from 70% to 100% with a more robust limiter or gradient clipping strategy.
5. **Complete all cases:** Run the three NACA laminar Re 5000 cases and the steady cylinder Re 20 case.
6. **Extend cylinder Re 200:** Run to $t = 300$ and extract Strouhal number, mean C_D , and lift amplitude from the periodic regime.
7. **Parallel validation:** Run systematic comparisons at 1, 2, 4, 8 ranks with timing data and load-balance analysis.

References

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- [2] J. Blazek, *Computational Fluid Dynamics: Principles and Applications*, 3rd ed. Butterworth-Heinemann, 2015.
- [3] G. Karypis and V. Kumar, “A fast and high quality multilevel scheme for partitioning irregular graphs,” *SIAM Journal on Scientific Computing*, vol. 20, no. 1, pp. 359–392, 1998.